Class 15: History of Quantum Mechanics – 2

In the last class we looked at the origin of Quantum Mechanics. Max Planck's investigation of blackbody radiation, and his unexpected discovery that matter exchanged energy with radiation in discrete increments of $h\upsilon$, which he called quanta, represents the start of quantum mechanics as a field of science. At that time Max Planck was unsure what his discovery implied, or whether nature actually behaved this way, but it led to a good fit with the data obtained from blackbodies.

It is of interest to note that if at the end of Planck's analysis, h had turned out to be zero, then it would have implied that matter exchanged energy with radiation in a continuous manner at all frequencies. Quantum mechanics would not have existed. The small, but non zero value of h is the signature that nature follows quantum mechanics.

Max Plank's discovery, helped in the explanation of not only blackbody radiation, but also of the other major experiments that had eluded satisfactory explanation up until that time.

The photoelectric effect displayed the following features:

1) No electrons are ejected regardless of intensity of the incident radiation, unless the frequency υ, of the radiation, exceeds a threshold value.

2) Kinetic energy of the ejected electrons increases with frequency υ of incident radiation and is independent of the intensity of the radiation

3) Even at low intensities electrons are ejected if the threshold frequency υ is exceeded.

To explain these observations, Einstein proposed the following equation:

$$\frac{1}{2}m_e v^2 = h\upsilon - \phi$$

where $\frac{1}{2}m_e v^2$ is the kinetic energy of the ejected electrons and ϕ is the work function of the solid from which the electrons are being ejected.

Einstein suggested that the energy of the incident light, at any given frequency υ, was given by the expression $h\upsilon$ - an idea that he borrowed and extended from Plank's analysis of blackbody radiation. Einstien's equation, which successfully explained the data corresponding to the photoelectric effect, implied the following:

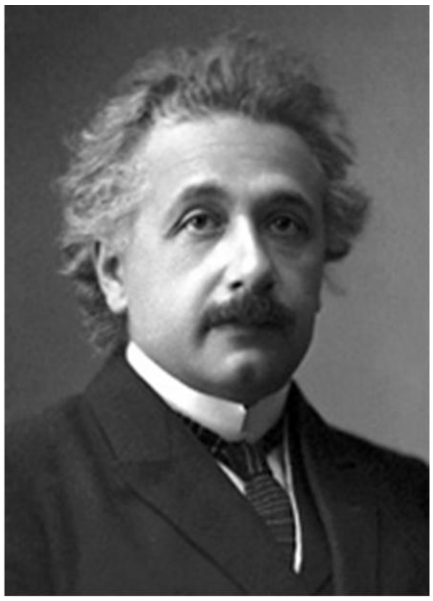
1) Electromagnetic radiation of frequency υ cannot possess any arbitrary amount of energy – it can only possess energies $h\upsilon$, $2h\upsilon$, $3h\upsilon$, …, $nh\upsilon$.

2) Electromagnetic radiation of frequency υ behaves as though it consists of 1, 2, 3, …, n particles each with energy $h\upsilon$.

Specifically the equation implied that at a frequency υ, light could not have possess energy that were non integral multiples of $h\upsilon$. For example an energy of $1.25h\upsilon$ was prohibited at υ.

Light, which had been thought of as exhibiting wave like behavior for a very long time, suddenly seemed to display particle like behavior. Einstein called these particles of light as 'photons'. It is interesting to note that nearly 200 years before Einstein's analysis, Newton had suggested that light consisted of particles, which he called 'corpuscles'. However, when experiments showed that light displayed diffraction and interference phenomena, Newton's idea of particles of light was abandoned and the wave nature of light was 'established'. With Einstein's analysis, the idea of light as particles, resurfaced and the ideas had come through a full circle. Although Newton preceded Einstein in the idea of light as particles, Einstein takes the credit for this idea, since he placed it on a stronger theoretical footing and successfully explained phenomena on this basis, while Newton had merely suggested the possibility.

For his contribution to explaining the photoelectric effect, Einstein was awarded the Nobel Prize in 1921. A photograph of Einstein, and his citation is indicated in Figure 15.1 below.



Albert Einstein

The Nobel Prize in Physics 1921

"for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

http://nobelprize.org/nobel_prizes/physics/laureates/1921/index.html

Figure 15.1: Photograph of Einstein, and the citation mentioned as part of the Nobel Prize awarded to him.

Einstein was also responsible for introducing the world to the ideas of relativity, the general theory of relativity and the special theory of relativity. But Einstein was awarded the Nobel Prize for his work on the photoelectric effect, an indication perhaps of how significantly quantum mechanics changed the landscape of science.

While Einstein suggested that light, which had been thought of as waves, could also be thought of as particles, Louise de Brouglie considered the opposite – that particles could be thought of as waves.

Louis de Broglie proposed, in 1924, the idea that any particle, travelling with a linear momentum 'p', can be thought of as having a wavelength λ given by:

$$\lambda = \frac{h}{p}$$

This possibility that particles could be thought of as waves, was successfully explored by Davisson and Gremer. In 1927 they demonstrated that a beam of electrons could diffract. Electrons from a heated filament, incident on a Ni sample, demonstrated diffraction. There was an element of luck in their work since the wavelengths involved in their experiment worked out just right for them to observe the diffraction easily in reflection mode.

Louise de Broglie's theory suggested that even large, day-to-day, objects could be thought of has having a wavelength associated with them. Using his theory, we find that indeed such a wavelength can be determined even for large day-to-day objects, but it is just that the wavelength turns out to be insignificant in the size scale of day-to-day objects and hence does not impact our physical observation and interactions with these objects.

Between Einstein, Louis de Broglie, and Davison and Gremer, the world was introduced to the ideas that waves could show particle like behavior and all particles could show wave like behavior.

Quantum Mechanics presented within its framework the idea of wave-particle duality.
The approach presently adopted in physics is to treat matter either as waves or as particles entirely based on circumstances. Wherever convenient particle like description of matter is assumed, and wherever convenient wave like description of matter is used.

For his insights into the wave nature of particles, Louis de Broglie was awarded the Nobel Prize in 1929. Figure 15.2 below shows a photograph of Louis de Broglie, and the citation for his Nobel Prize.



Prince Louis-Victor Pierre Raymond de Broglie

The Nobel Prize in Physics 1929

"for his discovery of the wave nature of electrons"

http://nobelprize.org/nobel_prizes/physics/laureates/1929/index.html

Figure 15.2: Photograph of Louis de Broglie, and the citation mentioned as part of the Nobel Prize awarded to him.

New tools were required to deal with ideas of quantum mechanics. The trajectory of particles in classical physics, needed to fit inside the description of a wave, a requirement that was successfully addressed by Erwin Schrödinger.

Quantum mechanics accepts the wave-particle duality by treating the trajectory of a particle as a wave, represented by a wave function ψ. The wave function ψ, has the properties of the system and can be obtained by solving the Schrödinger wave equation:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi = E\psi$$

The Schrödinger wave equation *cannot be derived* from more fundamental principles. It is itself considered the fundamental principle. It merely states that the total energy of the system consists of the kinetic energy and potential energy of the system.

In the studies involving quantum mechanics, we typically identify the constrains placed on a system and solve the Schrödinger wave equation consistent with these constraints to obtain the wave function ψ of the system. This wave function ψ then encapsulates the properties of the system. Schrödinger provided us with the tool using which we could extract an understanding of the system within the framework of quantum mechanics. For his contributions to quantum mechanics Schrödinger was awarded the Nobel Prize in 1933. He shared that year's Nobel Prize with Paul Dirac, whose contributions we will

discuss a little later. Figure 15.3 below shows photographs of Schrödinger and Dirac and the citation for their Nobel Prize.



Erwin Schrödinger



Paul Adrien Maurice Dirac

The Nobel Prize in Physics 1933

"for the discovery of new productive forms of atomic theory"

http://nobelprize.org/nobel_prizes/physics/laureates/1933/index.html

Figure 15.3: Photographs of Erwin Schrödinger and Paul Dirac, and the citation mentioned as part of the Nobel Prize awarded to them.

The Schrödinger wave equation is easy to solve only for a few simple cases. In general it can get quite complicated to solve and may require specific mathematical tools in some cases.

The Schrödinger wave equation indicated above is one form of the equation, known as the time independent Schrödinger wave equation. There is another form of the equation called the time dependent Schrödinger wave equation. This later equation is in conflict with aspects of relativity and is yet another indication that there may be more to discover about the workings of nature.

While the Schrödinger wave equation became accepted as a fundamental aspect of the quantum mechanical world, considerable confusion prevailed on the significance of the wave function ψ. Max Born provided the interpretation of the wave function ψ. If ψ^* is the complex conjugate of ψ, he stated that $\psi\ \psi^* dx$, or $|\psi|^2 dx$, is the probability of finding the electron between x and x + dx. For this contribution to quantum mechanics,

called the Born interpretation, Max Born was awarded the Nobel Prize in 1954. A photograph of Born, and the citation for his Nobel Prize, are shown in Figure 15.4 below.



Max Born

The Nobel Prize in Physics 1954 *(shared)*

"for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wavefunction"

http://nobelprize.org/nobel_prizes/physics/laureates/1954/index.html

Figure 15.4: A photograph of Max Born, and the citation for his Nobel Prize

For a bound electron, for example, the wave function ψ will turn out such that $\psi\psi^* dx$ will have a high value in the vicinity of the atom, and will be virtually zero everywhere else.

Niels Bohr looked at atomic and molecular spectra and attempted to explain the discrete nature of these spectra. He proposed that electrons went around the nucleus in fixed orbits and that the energy released when electrons jumped from one orbit to another, was a fixed value depending on the orbits between which the jump occurred. While this planetary model of the atom has not turned out to be exactly right, it did explain the experimental spectra seen. Schrödinger showed that the results obtained using his wave functions were consistent with the predictions of Niels Bohr.

For his contributions to explaining atomic and molecular spectra, Niels Bohr was awarded the Nobel Prize in 1922. A photograph of Niels Bohr and the citation for his Nobel Prize, are shown in Figure 15.5 below.



Niels Henrik David Bohr

The Nobel Prize in Physics 1922

"for his services in the investigation of the structure of atoms and of the radiation emanating from them"

http://nobelprize.org/nobel_prizes/physics/laureates/1922/index.html

Figure 15.5: A photograph of Niels Bohr and the citation for his Nobel Prize.

As indicated earlier, it is not easy to solve the Schrödinger wave equation in many cases. The wave function itself may end up being very complicated. If the wave function can be thought of as a sum of many waves, then each measurement of momentum of the particle will result in a value corresponding to any one of the waves that go to make up the wave function. This idea and its surprising implication was explored by Heisenberg and led to his identification of the 'Uncertainty Principle'.

The 'uncertainty principle', attributed to Heisenberg, effectively states that if the location of a particle is identified very precisely, then determining its momentum will be very difficult or will be very imprecise. High school texts, in order to simplify the complexity involved, explain the theory by saying that to determine the position of a particle very precisely, we need to shine a light on the particle to see it. In this process the light impinges on the particle and it disturbs it. As a result the momentum of the particle changes and we are hence unable to determine the exact position as well as the exact momentum of the particle at the same time.

The problem with this explanation is that it conveys the sense that the Heisenberg uncertainty principle is just an experimental limitation. If it were only an experimental limitation, the situation should have likely improved over the years – we should have much lower uncertainty now, than nearly a hundred years ago, in view of our improved experimental facilities. However, this 'improvement' does not seem to have occurred. We don't hear news reports of lowered uncertainty with respect to the Heisenberg's

uncertainty principle. Such a lack of improvement is a hint that this principle is perhaps not one of experimental limitation.

A more advanced view of the uncertainty is as follows. Momentum and position of a particle are related to each other through a Fourier transform, and hence are referred to as conjugate variables of each other. It turns out that when we carry out Fourier transforms, then as a direct result of the Fourier transform process, the product of the variability of one variable and that of its conjugate variable is equal to or greater than a certain value.

For example, if A and B are variables related by a Fourier transform, and ΔA and ΔB are represent the variability associated with each of these variables, then:

$$\Delta A \times \Delta B \geq C$$

Where C is a constant

Any conjugate variable pair will therefore create the situation that when we try to decrease the variability in one variable, we will automatically increase the variability in the other. Therefore, this rule appears as a result of a mathematical necessity arising from the Fourier transform process. It has no direct link to experimental limitations.

The momentum p, and the position x, of a particle are conjugate variables. Therefore, Heisenberg proposed that if p is the momentum associated with a particle and Δp is the uncertainty in determining this momentum, and if x is the location of the particle and Δx is the uncertainty in determining its location, then the 'Uncertainty Principle' states:

$$\Delta p \times \Delta x \geq h/4\pi$$

One way to understand the situation is that many waves have to be added to get a description that ensures that the particle is confined to a small location. Therefore the particle can now have many wavelengths and hence many possible momenta.

For his discovery of the uncertainty principle' Heisenberg was awarded the Nobel Prize in 1932. Figure 15.6 below shows a photograph of Heisenberg and the citation for his Nobel Prize.



Werner Karl Heisenberg

The Nobel Prize in Physics 1932

"for the creation of quantum mechanics, the application of which has, inter alia, led to the discovery of the allotropic forms of hydrogen"

http://nobelprize.org/nobel_prizes/physics/laureates/1932/index.html

Figure 15.6: Photograph of Werner Heisenberg and the citation for his Nobel Prize.

The concept of the uncertainty principle is summarized in Figure 15.7 below.

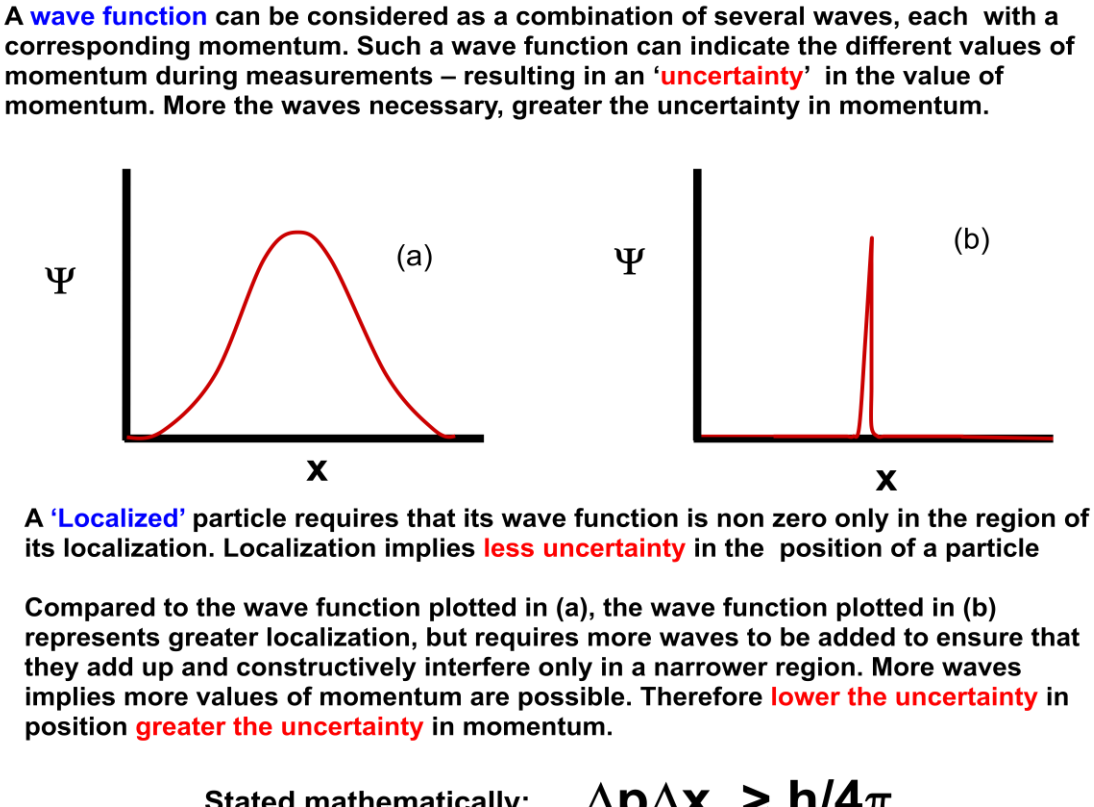


Figure 15.7: The concept of the uncertainty principle.

Specifically with respect to electrons, Wolfgang Pauli proposed that no two electrons can have all of their quantum numbers being identical. This is referred to as the Pauli's exclusion principle, and resulted in a Nobel Prize for Pauli in the year 1945. A photograph of Pauli and the citation for his Nobel Prize, are shown in Figure 15.8 below.



Wolfgang Pauli

The Nobel Prize in Physics 1945

"for the discovery of the Exclusion Principle, also called the Pauli Principle"

http://nobelprize.org/nobel_prizes/physics/laureates/1945/index.html

Figure 15.8: Photograph of Wolfgang Pauli and the citation for his Nobel Prize.

The key concepts of quantum mechanics and the people credited with their discovery, are summarized in Figure 15.9 below.

$E = h\upsilon$	Max Planck
$\frac{1}{2}m_e v^2 = h\upsilon - \phi$	Einstein
$\lambda = \frac{h}{p}$	Louis de Broglie
$\left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi = E\psi$	Schrödinger
$\psi\,\psi^* dx$, or $\lvert\psi\rvert^2 dx$	Born
$\Delta p \times \Delta x \geq h/4\pi$	Heisenberg

Figure 15.9: The key concepts of quantum mechanics, and the people credited with their discovery.

It is relevant to note that several of the ideas associated with quantum mechanics, resulted in Nobel Prizes for their proponents. Such is the significance of these discoveries. At the same time, although several major unexplained phenomena were successfully explained using the idea of quantum mechanics, there was considerable discomfort with the idea.

There was a school of thought that perhaps there was a deeper theory that did not require quantization but which would still explain the phenomena observed. Such thoughts did not materialize in all of the investigations since, and quantum mechanics came to stay. So when we sometimes face difficulties in understanding and accepting the ideas of quantum mechanics, when we feel that these ideas are not intuitive, we can take comfort from the fact that some of the greatest minds in the history of science also struggled to accept it and worried about its significance.